Area of Parallelograms

Lesson 5-1

Name: ____________ Date: ____________ Class: ____________

Key Vocabulary

Level 1 support

Picture first, then the word, then a plain-language meaning. Say each word out loud.

Think of a leaning door — the top and bottom are parallel, and the two sides are parallel, like a slanted rectangle

Parallelogram

A four-sided shape with two pairs of parallel sides.

If the bottom of a parallelogram is 10 cm, then b = 10 cm in the formula $A = b \times h$

Base

A side of the shape you use to find the area.

A dashed vertical line from the top side straight down to the base, forming a 90° angle — NOT the slanted side

Height

The straight-up distance from the base to the top.

A 3 cm × 4 cm rectangle covers 12 square centimeters — imagine 12 tiny 1×1 squares inside it

Area

How much space is inside a flat shape.

An L-shaped room = a 10×8 rectangle joined to a 4×5 rectangle; total area = 80 + 20 = 100 sq ft

Composite figure

A shape made by putting two or more simple shapes together.

$A = b \times h$ means Area equals base times height; for a parallelogram with $b = 6$ and $h = 4$, $A = 24$

Formula

A math rule written with symbols.

Key Ideas & Notes

🎯 **WHAT WE'RE LEARNING TODAY**

I can find the area of a parallelogram using base × height.

1 Notice

Your architecture firm is designing a parallelogram-shaped patio for a client's backyard.

2 Key idea

I can find the area of a parallelogram using base × height.

3 Words I'll use

Parallelogram **Base** **Height** **Area** **Composite figure** **Formula**

🤔 Think About It

- What shape is the patio?
- What measurements are given?
- How is a parallelogram different from a rectangle?

👀 Watch

Watch your teacher model one example. Jot what you see.

____

🤝 We try

Solve the next one together as a class.

____

____

✏️ You try

Now try one on your own.

____

____

Turn & Talk — Discussion Points

Level 1 support | Level 2

Talk with a partner about each prompt. If you need help getting started, use the Level 1 sentence stems. Ready for more? Try the Level 2 question.

LAUNCH

Look at the patio blueprint with a base of 14 feet and a height of 9 feet. Why do we use the height (9 ft) and not the slanted side to find the area?

Level 1 support

Start here: We use the height because it is perpendicular to the base.

💡 **Need a hint?**

Sentence starters (optional)

We use the height because it is ____ to the base.

Usamos la altura porque es ____ a la base.

The slanted side is not the height because ____.

El lado inclinado no es la altura porque ____.

Word bank: base | height | area | parallelogram | formula

Listen for: *Students connect 'height' to the perpendicular (straight-up) distance from base to top, and recognize the slanted side is longer and would overstate the area.*

Level 2

If the client tilted the patio to make it more slanted but kept the base at 14 ft and the height at 9 ft, would the area change? Justify your answer.

- The area would ____ because the formula only depends on ____ and ____.
- Tilting the shape changes the slanted side but not the ____, so ____.

EXPLORE

On the grid you cut a triangle off one end of the parallelogram and slid it to the other side. What shape did you make, and why does that prove the area is base x height?

Level 1 support

Start here: Rearranging the parallelogram makes a rectangle with the same base and height.

💡 **Need a hint?**

Sentence starters (optional)

When I move the triangle, I get a ____.

Cuando muevo el triángulo, obtengo un ____.

Area equals base times ____, so 14 times 9 equals ____.

El área es la base por ____, entonces 14 por 9 es ____.

It is like a rectangle because ____.

Es como un rectángulo porque ____.

Word bank: rearrange | rectangle | base | height | area

Listen for: *A strong answer names a rectangle, states the area is 14 x 9 = 126 sq ft, and explains the rearranged rectangle has the same base and height as the parallelogram.*

Level 2

A classmate says any quadrilateral's area is base x height. Critique this claim using what you know about parallelograms versus other four-sided shapes.

- This claim is wrong because ____ only works when ____.
- A counterexample is ____, which shows that ____.

CONNECT

The landscape architect tiles an 18 ft by 10 ft parallelogram courtyard and each tile covers 2 sq ft. Talk through how you find the number of tiles.

Level 1 support

Start here: First find the area, then divide by the tile size.

💡 **Need a hint?**

Sentence starters (optional)

First I find the area: 18 times 10 equals ___ square feet.

Primero hallo el área: 18 por 10 es ___ pies cuadrados.

Then I divide by 2 because ___, so she needs ___ tiles.

Luego divido entre 2 porque ___, entonces necesita ___ baldosas.

Word bank: area | base | height | parallelogram | tiles

Listen for: *Students compute area = 180 sq ft, then divide 180 / 2 = 90 tiles, and explain why dividing by the tile size gives the count.*

Level 2

Tiles come in boxes of 12. Generalize a rule for finding how many full boxes the architect must buy for any courtyard area, and apply it here.

- A rule for boxes is ___ because ___.
- For this courtyard she needs ___ boxes, since ___.

Guided Examples

Example 1

What is the area of a parallelogram with base 10 cm and height 7 cm?

Solution: $A = b \times h = 10 \times 7 = 70$ square centimeters.

Answer: A. 70 sq cm

Example 2

A parallelogram has an area of 54 sq ft and a base of 9 ft. What is the height?

Solution: $A = b \times h \rightarrow 54 = 9 \times h \rightarrow h = 54 \div 9 = 6$ ft.

Answer: A. 6 ft

Example 3

What is the area of a parallelogram with base 9 cm and height 5 cm?

Solution: $A = b \times h = 9 \times 5 = 45$ square centimeters.

Answer: A. 45 sq cm

Write About the Math

The Writing Revolution

I can explain how I found the area using the words base, height, area, and formula.

1. Kernel Sentence

subject + verb

Model: Parallelogram is a four-sided shape with two pairs of parallel sides.
Paralelogramo es una figura de cuatro lados con dos pares de lados paralelos.

Write a kernel sentence about parallelogram. Use a subject and a verb.
Escribe una oración base sobre paralelogramo. Usa un sujeto y un verbo.

____

____

2. Sentence Expansion

because · but · so

Kernel: Parallelogram matters in math
Paralelogramo importa en matemáticas

Expand the kernel three ways. Add a reason, a contrast, and a result.

because / ***porque***

Parallelogram matters in math because ____.
Paralelogramo importa en matemáticas porque ____.

____

but / ***pero***

Parallelogram matters in math, but ____.
Paralelogramo importa en matemáticas, pero ____.

____

so / ***entonces***

Parallelogram matters in math, so ____.
Paralelogramo importa en matemáticas, entonces ____.

____

3. Sentence Types

4 ways to write a math idea

Statement
Afirmación

Tell one true fact about parallelogram.
Di un hecho verdadero sobre parallelogram.

Parallelogram ____.

Question
Pregunta

Ask a question about parallelogram.
Haz una pregunta sobre parallelogram.

How does ____ ?
¿Cómo ____ ?

Exclamation
Exclamación

Show excitement about parallelogram.
Muestra entusiasmo sobre parallelogram.

Wow, ____ !
¡Guau, ____ !

Command
Mandato

Tell a partner what to do with parallelogram.
Dile a un compañero qué hacer con parallelogram.

First, ____ .
Primero, ____ .

4. Explain Your Reasoning

use a sentence starter

We use the height because it is ____ to the base.
Usamos la altura porque es ____ a la base.

The slanted side is not the height because ____.
El lado inclinado no es la altura porque ____.

When I move the triangle, I get a ____.
Cuando muevo el triángulo, obtengo un ____.

Try It

Solve on your own. Check the answer key when you are done.

1. What is the area of a parallelogram with base 9 cm and height 5 cm?

A. 45 sq cm

B. 14 sq cm

C. 22.5 sq cm

D. 90 sq cm

Show your work:

2. Which measurement is the height of a parallelogram?

A. The perpendicular distance between the base and the opposite side

B. The length of the slanted side

C. The perimeter divided by 4

D. The longest side

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A parallelogram has a base of 12 cm. If you double the height, the area doubles too. Explain why this happens using the formula, and give a specific example with numbers.

Sentence starter: When the height doubles from ____ to ____, the area changes from ____ to ____ because in the formula A = b × h, the base stays at ____ and ____.

Show your work:

Reflect — Exit Ticket

A parallelogram has a base of 13 inches and a height of 5 inches. What is its area?

A. 65 sq in
B. 36 sq in
C. 18 sq in
D. 65 in

Your answer:

Answer Key & Teacher Guide

1. **Try It 1:** A. 45 sq cm *— $A = b \times h = 9 \times 5 = 45$ square centimeters.*
2. **Try It 2:** A. The perpendicular distance between the base and the opposite side *— The height is always the perpendicular (straight up-and-down) distance between the base and the opposite side, not the slanted side.*
3. **Exit Ticket:** A. 65 sq in *— $A = b \times h = 13 \times 5 = 65$ square inches.*

Writing (TWR) — what to look for

- **Kernel sentence:** A complete sentence needs a subject and a verb. Example: Parallelogram is a four-sided shape with two pairs of parallel sides.
- **Expansion:** *because* gives a reason, *but* shows a contrast or exception, *so* shows a result. Answers vary; each must keep the kernel idea and add the correct kind of detail.
- **Sentence types:** Statement ends with a period, question with "?", exclamation with "!", and a command starts with an action verb (a "bossy" verb).